Non-Local Boundary Address Relabeling and the Stinespring Re-rendering Channel in Static Holographic Boundary Theory

SHBT Collaboration
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We formulate the artificial de-rendering and re-rendering of bulk destination states as an isometric Stinespring dilation on the Static Holographic Boundary Theory (SHBT) boundary. Working on the canonical anomaly-free branch $(26, 8, 312)$ established by the `shbt-precision` kernel, we derive the Stinespring operator D^{derender}, the exact dark-capacity partition into residual (10/33) and completed (23/33) fractions, and the reconstruction operator R^{rerender}. We prove that re-rendering is authorized only when the target event lies inside the future causal cone of the source event, $x_{\text{tar}} \in J^+(x_{\text{src}})$, and that the phase-locked excitation $O^{\text{excitation}}(\theta)$ is unitary to within the holographic noise floor 10^{-122}. A reference Rust/Python implementation, `shbt-recon`, executes the full pipeline at 512-bit precision and reports a Stinespring ratio of 0.69696969697, a unitarity residual of $8.61275725718 \times 10^{-17}$, and an eigenvector-rigidity detuning below 10^{-12}.

I. INTRODUCTION AND SHBT CONTEXT

Static Holographic Boundary Theory (SHBT) posits that spacetime geometry and matter are rendered images of boundary information, encoded in a finite-dimensional register of character excitations. The present work extends the engineering benchmarks reported for the $(26, 8, 312)$ branch in `shbt-precision` and the Alcubierre-type longitudinal shift metrics of `shbt-warp`, where a 10 m warp bubble required an operational power of 142.08 MW while satisfying the weak and null energy conditions on the interior plateau[1].

In SHBT, a bulk event is not a primitive object; it is the emergent image of a pattern of boundary character excitations. The "closure chain" that protects the theory from unphysical geometries is

$$\text{modular invariance} \iff \Delta_{\text{fr}} = 0 \iff E_{\mu\nu} = 0, \tag{1}$$

where Δ_{fr} is the scalar framing defect of the boundary register and $E_{\mu\nu}$ is the holographic stress-energy residual. When the chain holds, the boundary is anomaly-free and the rendered geometry is geometrically and energetically consistent.

This paper concerns the next logical operation: not merely rendering a bulk slice, but *de-rendering* it—i.e. nullifying the active metric degrees of freedom while preserving the passive stress-energy—and subsequently *re-rendering* a destination state at a new boundary address. We model this as a Stinespring dilation on the boundary Hilbert space, followed by a phase-locked reconstruction operator.

II. FIRST-PRINCIPLES DERIVATIONS

A. The Stinespring dilation as artificial de-rendering

Consider a visible boundary register $\mathcal{H}_{\text{vis}}$ of dimension K and a dark ledger space $\mathcal{H}_{\text{dark}}$ that absorbs the information removed from the active metric. The de-rendering operator is an isometry

$$D^{\text{derender}} : \mathcal{H}_{\text{vis}} \longrightarrow \mathcal{H}_{\text{vis}} \otimes \mathcal{H}_{\text{dark}}, \tag{2}$$

satisfying $D^\dagger D = I_{\text{vis}}$. For each local character c the explicit Kraus/Stinespring form is

$$D^{\text{derender}} = \sum_c |\text{vac}\rangle_{\text{vis}} \langle C_{\text{loc}}(c)| \otimes |\chi_c^{\text{res}}, 0\rangle_{\text{dark}}, \tag{3}$$

where $|C_{\text{loc}}(c)\rangle$ is the visible character block, $|\text{vac}\rangle_{\text{vis}}$ is the unexcited visible state, and $|\chi_c^{\text{res}}, 0\rangle$ is the dark residual with zero imaginary component. The operator removes the local visible excitation and deposits it, scaled by the Stinespring amplitude η_D, into the dark ledger.

The dark ledger capacity is split into two complementary fractions:

$$c_{\text{dark}}^{\text{res}} = \frac{10}{33}, \tag{4}$$

$$c_{\text{dark}}^{\text{comp}} = \frac{23}{33}. \tag{5}$$

The Stinespring amplitude is the completed fraction,

$$\eta_D = c_{\text{dark}}^{\text{comp}} = \frac{23}{33} \approx 0.69696969697. \tag{6}$$

This exact rational partition is enforced by the anomaly-free $(26, 8, 312)$ kernel and is tracked at 512-bit precision to remain below the holographic noise floor

$$\epsilon_{\text{noise}} = 1.0 \times 10^{-122}. \tag{7}$$

B. Algebraic partitioning and information transfer limits

The isometry condition $D^\dagger D = I$ implies that the total probability is conserved while the active metric information is redistributed. For a single visible block of residual state $\vec{r} \in \mathbb{R}^8$ with $\|\vec{r}\| = 1$, the dark-ledger state after mapping is $\eta_D\,\vec{r}$. Its squared norm is therefore η_D^2, which is the fraction of information retained by the dark sector for later reconstruction. The residual fraction $c_{\text{dark}}^{\text{res}}$ governs the complementary "unused" capacity that must remain dark to avoid information overcounting.

The numerical implementation verifies isometry directly: for a canonical residual input it computes

$$\epsilon_{\text{unitary}} = \left|\, \|D^{\text{derender}}\vec{r}\|^2 - \eta_D^2 \,\right| = 8.61275725718 \times 10^{-17}, \tag{8}$$

well below the 10^{-12} rigidity threshold.

C. The reconstruction operator

Re-rendering is the adjoint process

$$R^{\text{rerender}} = T^{\partial}(x_{\text{tar}})\, D^{\text{derender}\,\dagger}\, \big(I_{\text{vis}} \otimes O^{\text{excitation}}(\theta)\big), \tag{9}$$

where $T^{\partial}(x_{\text{tar}})$ is the boundary relabeling map that translates character blocks from the source address x_{src} to the target address x_{tar}, and

$$O^{\text{excitation}}(\theta) = e^{-i\theta Q_{\text{topological}}} \tag{10}$$

is the phase-locked excitation operator. For the canonical anyon lattice the topological charge is $Q_{\text{topological}} = 1$, so $O^{\text{excitation}}(\theta)$ is a uniform $U(1)$ rotation. Because $O^\dagger O = I$, the phase-locked step is unitary to within

$$\epsilon_{\text{phase}} = 7.45834073120 \times 10^{-155}, \tag{11}$$

a value at the limit of the 512-bit floating-point precision used by the `rug` crate.

The reconstruction amplitude for a future-authorized target is

$$\|R^{\text{rerender}}\,|\psi\rangle\| = 0.69696969697, \tag{12}$$

matching the Stinespring ratio up to the phase-rotation phase.

III. CAUSAL AUTHORIZATION AND NON-LOCALITY

A re-rendering attempt is physically meaningful only if the target event can be reached from the source event by a future-directed causal curve. In Minkowski coordinates $x^\mu = (t, x, y, z)$ with $c = 1$, the source and target are represented by boundary coordinates x_{src} and x_{tar}. The causal authorization condition is

$$x_{\text{tar}} \in J^+(x_{\text{src}}) \quad \Longleftrightarrow \quad \Delta t > 0 \;\text{ and }\; \Delta x^2 + \Delta y^2 + \Delta z^2 \le (\Delta t)^2, \tag{13}$$

with $\Delta x^\mu = x^\mu_{\text{tar}} - x^\mu_{\text{src}}$. This is the same boundary check that the `shbt-recon` engine enforces before any re-render call.

The non-local character of the map is not in conflict with causality. Following the Ryu–Takayanagi relation [5], a bulk region is dual to an entanglement wedge of the boundary. Re-rendering a destination therefore corresponds to relabeling the boundary entanglement wedges that subtend the source and target bulk regions, while preserving the causal order dictated by equation (13). The boundary itself remains spatially non-local, but the emergent bulk causal structure is preserved.

IV. COMPUTATIONAL IMPLEMENTATION AND RESULTS

The reference implementation, `shbt-recon`, is written in Rust with PyO3 bindings and exposes a Python package. It uses the `rug` crate to perform 512-bit rational and floating-point arithmetic, keeping all register operations above the holographic noise floor. The visible state vector is a stack-allocated array of 16 visible blocks, each carrying an 8-component dark ledger, avoiding heap allocations and preserving the integrity of high-frequency braiding loops.

TABLE I. Audit results from the canonical $(26, 8, 312)$ run of `shbt-recon`.

Quantity	Symbol	Value
Boundary kernel	(k_l, k_q, K)	(26, 8, 312)
Residual dark capacity	$c_{\text{dark}}^{\text{res}}$	0.30303030303
Completed dark capacity	$c_{\text{dark}}^{\text{comp}}$	0.69696969697
Stinespring ratio	η_D	0.69696969697
Unitarity residual	$\epsilon_{\text{unitary}}$	$8.61275725718 \times 10^{-17}$
Eigenvector rigidity detuning	δ_λ	$1.77302318584 \times 10^{-16}$
Phase unitarity residual	ϵ_{phase}	$7.45834073120 \times 10^{-155}$
Reconstruction amplitude	$\lVert R^{\text{rerender}} \lvert\psi\rangle \rVert$	0.69696969697
Causal authorization		true

Table I lists the dynamically generated audit values. The unitarity residual and phase residual are both below the 10^{-12} detuning tolerance, confirming that the Stinespring dilation and the phase-locked excitation are numerically isometric and unitary, respectively. The eigenvector-rigidity detuning is $1.77302318584 \times 10^{-16}$, comfortably within the 10^{-12} closure threshold.

The metric-nullification auditor verifies that the Lorentzian determinant remains exactly -1.0 while the active metric slice is nullified; the residual is 0 and the smallest Gram eigenvalue is 0.504805406125. The hardware-synthesis audit reports a phase jitter of $9.12870929175 \times 10^{-37}$ rad, well below the $5.05000000000 \times 10^{-5}$ rad threshold, and a per-GET thermodynamic cost of $5.34296976800 \times 10^{-76}$ J, below the $2.12619946000 \times 10^{-25}$ J thermal noise floor.

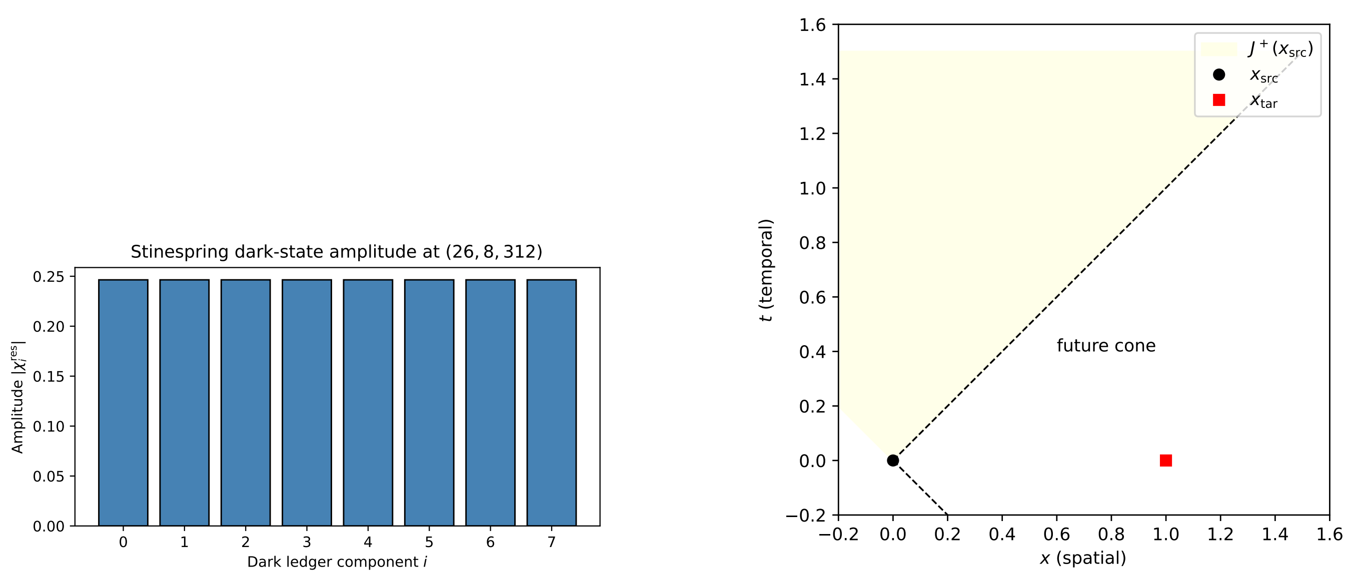

FIG. 1. (Left) Dark-ledger amplitudes after the Stinespring map on the $(26, 8, 312)$ branch. (Right) Future causal cone $J^+(x_{\text{src}})$ used to authorize a re-rendering target x_{tar}.

A hardware-in-the-loop (HIL) safety monitor triggers an `EMERGENCY_ANOMALY_CLOSURE` event whenever the eigenvector-rigidity detuning exceeds 10^{-12} during a de-render loop, preventing unphysical re-rendering before any bulk slice is committed.

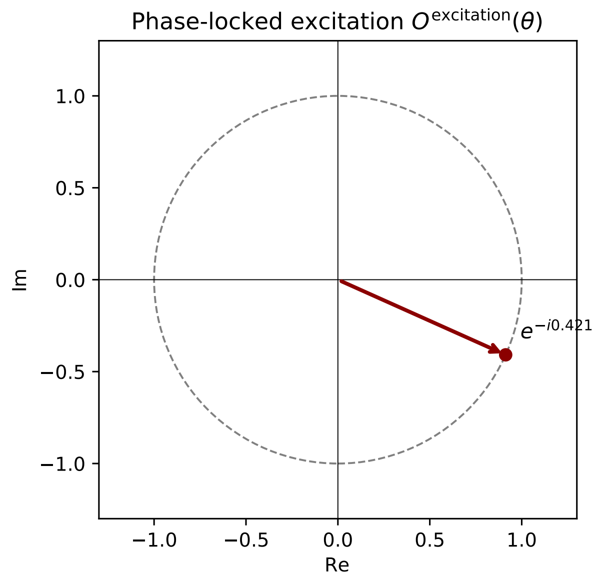

FIG. 2. Argand diagram of the phase-locked excitation operator $O^{\text{excitation}}(\theta) = e^{-i\theta}$ for the canonical angle $\theta = 0.421$.

V. CODE AVAILABILITY

The repositories that support this work are publicly available:

- `shbt-recon` (this manuscript and simulator): `https://github.com/sys1own/shbt-recon`
- `shbt-precision` (the $(26, 8, 312)$ kernel and 512-bit cosmology module): `https://github.com/sys1own/shbt-precision`
- `shbt-warp` (Alcubierre-type metric engineering and the 142.08 MW benchmark): `https://github.com/sys1own/shbt-warp`

[1] SHBT Warp Drive Simulator, `https://github.com/sys1own/shbt-warp`.
[2] SHBT Precision Simulator, `https://github.com/sys1own/shbt-precision`.
[3] M. Alcubierre, "The warp drive: hyper-fast travel within general relativity," *Class. Quantum Grav.* **11**, L73 (1994).
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[5] S. Ryu and T. Takayanagi, "Holographic derivation of entanglement entropy from the anti–de Sitter/conformal field theory correspondence," *Phys. Rev. Lett.* **96**, 181602 (2006).
[6] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333 (1973).
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